Math Intervention Decision Guide | For Leaders

For instructional coaches, MTSS leads and campus leaders: a Data-Based Individualization (DBI) protocol for when a student isn't responding to Tier 2 math intervention

Step 1: Look at the data, then look behind it

Start with the progress-monitoring trend, but never make an intervention or referral decision on scores alone. Check these four things first:

Check	Ask yourself
Fidelity	Was the intervention delivered as designed, including the manipulatives and modeling?
Dosage	Did the student get the planned sessions? Were missed sessions made up?

Check	Ask yourself
Access	Is language or reading load hiding what the student knows in math?
Data quality	Do the probes measure the same skill at the same difficulty? Is the goal reasonable? Are prerequisites in place?

Step 2: Talk with the teacher (coaching, not evaluating)

Area	Questions to ask
Instruction	"Can you walk me through what a typical 30-minute intervention session looks like?"
Fidelity	"Which parts of the lessons feel comfortable, and which are hard to implement?"
Math needs	"Is the struggle identifying fractions, equivalence, operations, or all three?"
Language	"How does the student do when you read the problem aloud or use a visual model?"
Attendance	"Were missed sessions made up? Has the student received the full planned dosage?"
Monitoring	"Are the weekly probes measuring the same skills at a similar difficulty?"
Student voice	"What does the student say is confusing? When do they show confidence?"

Also gather: recent work samples and a short observation of an intervention session.
Goal of the conversation: decide whether the intervention addresses the student's actual skill gap, whether it's delivered as designed, and whether language or other barriers block the student's access. *It may be more than one.*

Step 3: Adjust in this order

I Strengthen implementation
- Use the lessons as designed: concrete models, explicit modeling, guided practice, corrective feedback
- Name the essential components and track how consistently they're delivered

▶

II Remove language barriers
- Partner with the multilingual learner specialist
- Visual vocabulary, explicit math language, explaining with diagrams and manipulatives
- Keep grade-level math expectations

▶

III Monitor and review
- Weekly comparable probes
- Log attendance and fidelity
- Compare the trend to the goal line
- Review in 4–6 instructional weeks

Step 4: Decide at the review meeting

On track
Trend meets or beats the goal line. Continue, then plan to fade support.

Some growth, not enough
Study the student's wrong answers for patterns (for example, adding denominators), then change one thing at a time: time, group size, CRA stage or format.

Little to no response
Even with strong fidelity and full dosage, intensify to Tier 3: a smaller group, more time, more individualized instruction.

Systems check: is the campus set up for intervention to work?

- □ Intervention time is protected in the master schedule
- □ Interventionists have materials, manipulatives and training
- □ Fidelity data is collected, not just student scores
- □ Probes and decision rules are the same across classrooms
- □ MTSS meetings run on a set schedule with data in hand
- □ Multilingual and special education staff join decisions

Step 5: Coach the teacher with a short cycle

1 Validate
Start with a strength you saw in the session and what the teacher noticed about the student.

▶

2 Model
Teach part of a lesson with fraction strips, a think-aloud and guided practice.

▶

3 Co-plan & rehearse
Plan the next lesson together; the teacher practices with the manipulatives and math vocabulary.

▶

4 Observe & give feedback
Watch a short part of the session and name the specific moves that worked.

▶

5 Review data together
Look at work samples and the progress-monitoring trend.

The larger goal: build the teacher's capacity so these practices help every student who needs math intervention, not just one.

Example: comparing 1/2 and 2/3 using the CRA sequence

Instead of ten worksheet problems, move the student through each stage and ask them to explain which is greater and why.

Concrete	Representational	Abstract
Line up fraction strips for 1/2 and 2/3. Which is longer?	Draw both on a number line or as bar models.	Find common denominators: 3/6 < 4/6, so 1/2 < 2/3.

Observation look-fors

- ☐ Session happened as scheduled (time and length)
- ☐ Used concrete models or visuals
- ☐ Explicitly modeled the skill with a think-aloud
- ☐ Guided practice with corrective feedback
- ☐ Student explained their thinking aloud
- ☐ Pre-taught or reviewed key math vocabulary
- ☐ Independent practice on the target skill
- ☐ Progress probe given this week (same form/level)

Special education referral: guardrails

- **Don't refer automatically** because of a low screener score or one missed intervention goal.
- **Don't make the student wait** for another intervention cycle when a disability is already suspected.
- **Consider referral now** if evidence suggests a disability: persistent difficulty across settings, a history of significant math deficits, and concerns not explained by language acquisition or lack of appropriate instruction.
- **If not yet**, strengthen the intervention, gather diagnostic information and keep monitoring.

Under IDEA: MTSS cannot delay or deny an evaluation when a disability is suspected. Parents may request an initial evaluation at any time. Limited English proficiency cannot be the determining factor for eligibility.

Step 6: Share the recommendation at the MTSS meeting

Sample script you can adapt:

"The data show a persistent need in math, particularly fractions. Before attributing limited progress to a disability, I want to understand whether the intervention has been implemented consistently and whether academic language is affecting the student's access to instruction. I recommend reviewing diagnostic errors, strengthening explicit instruction and visual representations, supporting the teacher through a short coaching cycle, and continuing consistent progress monitoring. At the same time, we should review the full body of evidence to determine whether a disability is suspected. If so, we move forward with an evaluation without waiting for another MTSS cycle. Our goal is to make decisions based on both student outcomes and the quality of instruction the student has received."

Remember: MTSS is for every student, including general education, emergent bilingual, 504 and special education students. It is not a pathway to a special education label; it is how we match instruction to need.

Adapted from the National Center on Intensive Intervention's Data-Based Individualization (DBI) process. Designed by Yvette Saucedo, M.Ed. For training use; scenarios are fictional and contain no student information.